VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:

Continuous Assessment Test (CAT) – II - March 2026

Programme	:	B.Tech (CSE & its Specialised Branches)	Semester	:	WS-2025-26
Course Code & Course Title	:	BCSE305L, Embedded Systems	Class Number	:	CH2025260502876 CH2025260502877 CH2025260502878 CH2025260502880 CH2025260502881 CH2025260503397
Faculty	:	Dr. G. Gugapriya, Dr. K. O. Mohammed Aarif, Dr. S.Suhasini , Dr. Sritama Roy, Dr. P. Latha, Dr. M. S. Bala Murugan	Slot	:	A1 + TA1
Duration	:	90 Minutes	Max. Mark	:	50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1	a	A networking company uses a compression technique to encode the message before transmitting over the network. Suppose the message contains the following characters with their frequency: **Character** / **Frequency**: A 5; B 9; C 12; D 13; E 16; F 45 If the compression technique used is Huffman Coding, how many bits will be saved in the message? (3 Marks) Also draw the respective Huffman coding tree. Note that each character in input message takes 1 byte. (4 marks)	7	2	K3
	b	A financial institution applied a compression technique to its encrypted transaction records and observed that the compression resulted in little or no reduction in file size. What is the reason for this outcome? What corrective measure should be taken to ensure effective compression?	3	2	K3
2		Create a Use Case Diagram for Course Management Software: 1) At the beginning of each semester,	10	3	K4

<table>
<tr><td></td><td></td><td>- Each professor shall register the courses that he is going to teach.
2) A student can select up to four-course offerings.
-During registration a student can request a course catalogue showing course offerings for the semester.
-Information about each course such as professor, department and prerequisites would be displayed.
- The registration system sends information to the billing system so the students can be billed for the semester.
3) For each semester, there is a period of time during which dropping of courses is permitted.
4) Professors must be able to access the system to see which students signed up for each of their course offerings</td><td></td><td></td><td></td></tr>
<tr><td rowspan="2">3</td><td>a</td><td>Using a suitable graphical asynchronous modeling tool, describe and analyze the concurrent processes involved in the Smart Parking System based on the following attributes:
• Places: Parking spaces, Entry/Exit gates, Payment kiosk
• Transitions: Car arrival, Car parking, Payment processing, Car departure
• Tokens: Cars, Parking tickets</td><td>5</td><td>3</td><td>K4</td></tr>
<tr><td>b</td><td>How would you model the sequence of interactions between a patient, doctor, and billing system when a patient requests an appointment and pays the consultation fee?</td><td>5</td><td>3</td><td>K3</td></tr>
<tr><td>4</td><td></td><td>Table I presents an RMS schedule for a real-time military tracking system. The system consists of 3 tasks to be scheduled, each with their corresponding computation time and period.
<u>Table I: Task Scheduling Requirements</u>
<table><tr><td>Task</td><td>Description</td><td>Computation Time (C)</td><td>Period(T)</td><td>Priority</td></tr><tr><td>UTL</td><td>Update Track Log</td><td>10</td><td>20</td><td></td></tr><tr><td>UTr</td><td>Object Tracking</td><td>3</td><td>10</td><td></td></tr><tr><td>GPST</td><td>GPS Triangulation</td><td>1</td><td>5</td><td></td></tr></table>Verify if the schedulability test for RMS was correct by calculating the response times of all three tasks in Table I. (7 Marks)
Fill the "Priority" column as necessary. (1 Mark)
State your findings and compare it with the schedulability test. (2 Marks)</td><td>10</td><td>4</td><td>K4</td></tr>
<tr><td>5</td><td></td><td>Consider that the RMS schedule in Table I needs to be converted into an EDF schedule, with arrival times 1, 3, and 0 for tasks UTL, UTr, and GPST, respectively. Determine if this revised task set is schedulable using
(1) EDF Schedulability test (3 marks)
(2) Timing diagram method. (7 Marks)</td><td>10</td><td>4</td><td>K4</td></tr>
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**********All the best **********